

DHRUVAL ANANDKAR

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PROFESSIONAL SUMMARY

Computer Science (Honors) senior specializing in distributed systems, data engineering, and cloud-native architectures. Experienced in bridging the gap between complex data pipelines and enterprise reporting. Seeking Software Engineer (SWE), Backend Engineer, Data/Business Analyst, or AI-ML Engineer roles to leverage expertise in building event-driven MLOps infrastructure, MERN stack applications, and automated business intelligence dashboards.

EDUCATION

Bachelors in Computer Science, Honors Program Ashland University, Ashland, Ohio	December 2026 GPA: 3.7 / 4.0
Associate in Computer Engineering Parul University, Vadodara, Gujarat, India	May 2020 – May 2023 GPA: 4.0 / 4.0

RELEVANT COURSEWORK

Data Structures & Algorithms, Principles of Operating System, Database Systems, Distributed Computing, Machine Learning, Object-Oriented Design.

PROFESSIONAL EXPERIENCE

Ashland University – Ashland, Ohio, USA Data Engineering & Business Intelligence Intern	Oct 2025 – Present
- Engineered scalable Python backend pipelines and optimized SQL queries to integrate Snowflake with Microsoft Fabric, eliminating processing bottlenecks for downstream services - Developed automated data transformation scripts and strict Pydantic validation checks, reducing schema violations by 98% and ensuring high-fidelity data for enterprise analytics - Developed interactive Power BI dashboards for university financial and census analytics, managing data transformations in Power Query and resolving complex DAX calculations	
Ashland University – Ashland, Ohio, USA Research Team Lead – Data & Full-Stack Development	Oct 2024 – Present
- Directed the end-to-end development of an IoT platform, programming NodeMCU microcontrollers in C++ to capture and transmit real-time telemetry and sensor data - Engineered Python REST APIs and automated CI/CD deployment pipelines to sync live hardware telemetry with Firebase, achieving sub-200ms latency for machine learning predictions	
Smith Structure India Pvt. Ltd. – Kheda, Gujarat, India Development Intern – Internal Tools	Dec 2022 – Jan 2023
- Provided dedicated IT sector support, assisting the technical operations team with internal technology workflows and system troubleshooting - Maintained technical documentation and supported the daily management of internal IT infrastructure	
Gujarat State Electricity Corporation Limited (GSECL) - Wanakbori, Gujarat, India Systems Development Intern	Apr 2022 – May 2022
- Retrieved data from Oracle databases and documented IT infrastructure - Analyzed operational data using Advanced Excel to support the technical operations team	

COMPETENCIES

Functional Skills: Software Development Life Cycle (SDLC) • Backend Architecture • API Design • CI/CD Principles • Secure Environment Management • Production Debugging • Agile & SCRUM

Technical Skills:

- Languages:** Python, Java, JavaScript (ES6+), TypeScript, C++, SQL
- Frameworks & Libraries:** React.js, Node.js, Express.js, FastAPI, LangChain, LangGraph, Pydantic, Tailwind CSS
- Backend & Cloud:** AWS (EC2, S3), Docker, Git/Github, Microservices, REST APIs, CI/CD
- Data & ML:** PostgreSQL, MySQL, MongoDB, TimescaleDB, Redpanda (Kafka), Snowflake, Databricks, Microsoft Fabric, XGBoost, SciPy, ETL
- BI & Analytics:** Power BI, DAX, Power Query, Data Modeling, Data Visualization, Excel

TECHNICAL PROJECTS

- Vajra (MLOps Orchestrator):** Event-driven telemetry pipeline (**FastAPI, Kafka, TimescaleDB**) where a **LangGraph** agent detects data drift via KS-testing and autonomously retrains and promotes an **XGBoost** challenger model.
- AgriScience:** Built **Python/SQL** ETL pipelines aggregating distributed **IoT** sensor data for crop forecasting and trend analysis.
- Ashland Public Transit (Senior Year Thesis Project):** Engineered a fault-tolerant real-time GPS tracking system using **Node.js, Express, Socket.IO**, and **MongoDB**, with **JWT** role-based access control across rider, driver, dispatcher, and admin roles.
- JobGenie AI:** GPT-4 job match **SaaS** featuring AI document parsing to transform unstructured text into structured candidate profiles using **LangChain, React**, and **Stripe**.
- CortexLab:** Built a **Python** and **LangChain** multi-agent AI assistant to automate complex workflows and extract structured insights from unstructured natural language.
- News Bias Aggregator (Visualization):** Developed data visualization app to analyze multi-source scores, presenting clear findings to non-technical users.
- Vapi Outbound Campaign:** Built a real-time automation tool using the Vapi API for dynamic outbound call-flow logic and automated data handling.

CERTIFICATES & HONORS

Voice Assistant with OpenAI & IBM Watson	CognitiveClass.ai: Oct 2024
Data Science with Python	SoloLearn: Jun 2021
UN SDG Innovation & Entrepreneurship Challenge	Ashland University: Sep 2025
Containers & Kubernetes Essentials	CognitiveClass.ai: Jun 2026